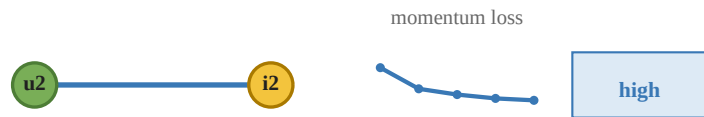


Loss-only view

Two interactions can exhibit similar learning difficulty

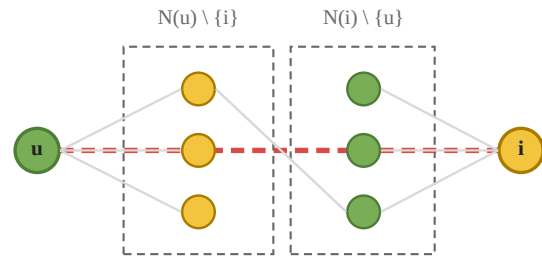


Noisy-like or hard-but-clean? Loss alone is ambiguous.

(a) Loss-only ambiguity

Structure--dynamics view

Bilateral context separates distinct meanings of high loss



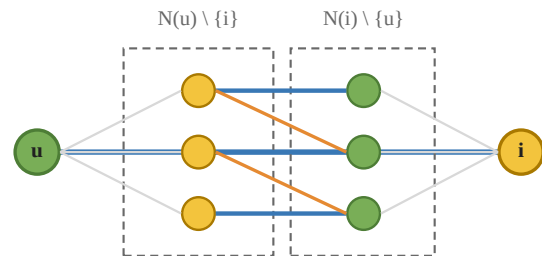
Noisy-like edge

high momentum risk

low user-side

low item-side

refine



Hard-but-clean edge

high momentum risk

high user-side

high item-side

retain

(b) Bilateral structural evidence